

PRODUCT REQUIREMENT DOCUMENTATION (PRD)

UMHackathon 2026



Project Name: StockFlowAI

Version: 1.0

Domain: 1

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System Title: AI Store Operations Copilot for Stock Directory and Supplier Invoice Processing

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1. Project Overview

- **Problem Statement:** Retail stores nowadays still rely on manual process flow to compare supplier invoices with received stock, report discrepancies, and escalate issues. Staff often use notes, spreadsheets, or chat messages, which causes repeated key-in work, inconsistent reporting, and delayed resolution. This matches the hackathon focus on fragmented and manual workflows that require human interpretation and repeated coordination
- **Target Domain:** AI Systems & Agentic Workflow Automation for retail store operations, focused on stock discrepancy handling and supplier invoice processing.
- **Proposed Solution Summary:** StockFlow AI is a workflow system that uses Z.AI GLM as the central reasoning engine to convert messy store inputs into structured discrepancy cases. The system accepts natural-language issue descriptions or invoice-related text, extracts key details, detects missing information, recommends workflow actions, and tracks case status. This directly aligns with the requirement that GLM be central to reasoning and workflow execution.

2. Background & Business Objective

- **Background of the Problem:** In store receiving operations, discrepancies between invoiced stock and actual received stock are common. Nowadays, these cases are often handled manually across different tools, making the process slow, error-prone, and hard to track.
- **Importance of Solving This Issue:** Improving this workflow can reduce manual effort, improve inventory accuracy, speed up issue handling, and create a clearer operational audit trail. This supports the hackathon goal of building reliable and intelligent workflow execution beyond simple chatbot behaviour.
- **Business Objective:** The objective of StoreFlow AI is to shorten the time needed to create and route discrepancy cases, improve consistency of store reporting, and reduce manual interpretation work through AI-assisted workflow automation.

3. Product Purpose

3.1 Main Goal of the System: StoreFlow AI helps store staff turn unstructured discrepancy reports into structured, actionable, and trackable operational cases. The system is designed as a **stateful workflow engine**, not just a chatbot.

3.2 Intended Users (Target Audience):

- Store staff / receiving staff
- Store manager / supervisor
- Inventory operations staff
- Finance reviewer

4. System Functionalities

4.1 Description: The system accepts free-text discrepancy input, interprets the issue, extracts relevant information, identifies missing fields, recommends the next workflow action, and generates a structured case output. This fulfills the requirement for unstructured input understanding, multi-step reasoning, and structured actionable outputs.

4.2 Key Functionalities:

- **Unstructured Case Intake:** Accepts discrepancy reports in natural language.
- **Case Classification:** Detects issue type such as invoice mismatch, stock shortage, over-delivery, or duplicate billing concern.
- **Field Extraction:** Extracts supplier, store, invoice number, product, expected quantity, actual quantity and discrepancy details.
- **Missing Information Detection:** Asks follow-up questions if required information is incomplete.
- **Workflow Recommendation:** Suggests next action such as manager review, finance review, or supplier dispute follow-up.
- **Structured Output Generation:** Produces a clear case summary and structured case record.
- **Workflow State Tracking:** Tracks case states such as NEW, PENDING_INFO, ROUTED, and RESOLVED.

4.3 AI Model & Prompt Design

4.3.1 Model Selection: The system uses **Z.AI GLM** as the main LLM because the hackathon requires GLM to act as the central reasoning engine for workflow understanding, decision-making, and structured output generation.

4.3.2 Prompting Strategy: StoreFlow AI uses a multi-step agentic prompting strategy:

1. Classify the case
2. Extract structured entities
3. Determine missing information
4. Propose workflow action
5. Return a structured output schema

This prompting strategy is chosen because the use case requires multiple decision points and the domain explicitly expects workflow-stage reasoning and orchestration rather than a static chatbot response.

4.3.3 Context & Input Handling: The system accepts short to medium-length text input and pasted invoice content. Oversized inputs will be truncated or chunked before model processing.

4.3.4 Fallback & Failure Behaviour: If the model output is unclear or low confidence, the system will mark fields as missing, ask for clarification, or escalate to human review instead of guessing.

5. User Stories & Use Cases

- **User Stories:**

- As a store staff member, I want to describe a discrepancy in my own words so that I do not need to manually fill every field.
- As a store manager, I want the system to summarize and classify discrepancy cases so that I can review them faster.
- As a finance reviewer, I want structured discrepancy details so that supplier follow-up is more consistent.

- **Main Use Cases:**

- **Supplier Invoice Mismatch:** User reports invoiced quantity differs from received stock; system extracts details and creates a case.

- **Incomplete Discrepancy Report:** User submits vague information; system requests missing fields before proceeding.
- **Duplicate Billing Concern:** User suspects duplicate invoice entry; system classifies and routes the case for review.

6. Features Included (Scope Definition)

- Natural-language discrepancy intake
- AI case classification
- Entity and field extraction
- Missing-information clarification
- Workflow recommendation and routing
- Structured case output generation
- Basic workflow state tracking

7. Features Not Included (Scope Control)

- Full ERP / POS integration
- Full supplier portal integration
- Advanced enterprise analytics dashboard
- Full production-grade approval hierarchy
- Complex OCR for highly scanned handwritten invoices

8. Assumptions & Constraints

8.1 Assumptions

- Users can provide a short text description of the issue
- Some invoice or stock information is available in text form
- Users can respond to clarification questions when details are missing

8.2 Constraints

- Z.AI GLM must be implemented as the core AI reasoning engine
- Cloud services are not provided by the hackathon, so the prototype should remain lightweight

- The MVP must stay focused enough to be completed within the preliminary round timeline

9. Risks & Questions Throughout Development

9.1 Incomplete User Input: Users may submit vague descriptions

- Mitigation: The system will detect missing fields and request clarification before routing the case.

9.2 Hallucinated AI Output: The model may generate uncertain values if input is unclear

- Mitigation: The system will validate structured outputs, avoid auto-filling uncertain fields, and escalate unclear cases for human.

9.3 Scope Expansion

- Mitigation: The MVP will stay limited to one workflow: stock discrepancy and supplier invoice issue processing.

9.4 External Dependency on Z.AI API

- Mitigation: The system will isolate AI calls behind a dedicated AI service layer so the workflow logic can still be tested independently.

Questions:

- What minimum fields are required before routing?
- Should manager review or finance review be the default first step?
- How should multi-item discrepancy cases be handled in the MVP?

10. Reference

FastAPI. (n.d.). *FastAPI*. <https://fastapi.tiangolo.com/>

Microsoft. (2026). Microsoft 365 Copilot [Large language model].
<https://www.microsoft.com/en-us/microsoft-365-copilot>

Streamlit. (n.d.). *Streamlit documentation*. <https://docs.streamlit.io/>

Z.AI. (n.d.). *GLM-5 overview*. <https://docs.z.ai/guides/llm/glm-5>